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论文前言撰写提示词 2.0

By 迪娜学姐 ©版权所有

分为以下三步完成：

- 1 前言框架拟定提示词；
- 2 文献综述调研提示词；
- 3 前言全文撰写提示词。

1 前言框架拟定提示词

你是一位[免疫学]领域的资深教授，发表过 100 篇 top 期刊论文，有丰富的学术论文撰写经验。以下是我即将投稿到 **Frontiers in Immunology** 期刊的论文标题和摘要部分,figure legends, 和该期刊的 Aims and Scope。

请你先仔细阅读我给出的资料，然后给我三种本论文 Introduction 部分的框架方案，并说明理由。

Inputs

<Title>

{{你的论文标题 IFN λ 1 is a STING-dependent mediator of DNA damage and induces immune activation in lung cancer}}

</Title>

<Abstract>

{{你的论文摘要 Introduction: The importance of the cGAS-STING pathway and type I interferon (IFN) in anti-tumor immunity has been widely studied. However, there is limited knowledge about the role of type III IFNs in cancer settings. Type III IFNs, comprising IFN λ 1-4, are opposite to type I IFN only expressed by a few cell types, including epithelial cells, and the receptor subunit IFNLR1, is equally only expressed on limited types of cells. Methods: Gene and protein expression of the cGAS-STING signaling pathway was characterized in a series of non-small cell lung cancer (NSCLC) cell lines. Herring-testis DNA stimulation and chemotherapy drugs (doxorubicin and cisplatin) were used to activate the cGAS-STING pathway, and the level of activation was determined by measuring changes in the transcriptomic profile as well as type I and III IFNs by ELISA. Re-expression of IFNLR1 on cancer cell lines was achieved using CRISPR activation (CRISPRa) followed by evaluating chemotherapy-induced apoptosis using flow cytometry assays. Results: STING was not broadly expressed across the NSCLC cell lines. Those cancer cell lines expressing all relevant factors supporting the cGAS-STING pathway secreted IFN λ following STING activation whereas only few of them expressed IFN β . Treatment with chemotherapy drugs likewise preferentially induced IFN λ , which was abrogated in CRISPR-Cas9 STING knock-out cells. Expression of IFNLR1 was found downregulated in the cancer cell lines compared to the benign epithelial cell line Nuli-1. Rescuing IFNLR1 expression by CRISPRa in multiple cancer

cell lines sensitization them to IFN λ -stimulation and resulted in significant reduction in cell viability. Conclusion: Downregulation of IFNLR1 can be an immune evasion mechanism developed by cancer cells to avoid responding to endogenous type III IFNs. Thus, rescuing IFNLR1 expression in NSCLC in conjunction to chemotherapy may potentially be harnessed to elevate the anti-tumoral responses.}}

</Abstract>

<Figure legend>

你的论文 figure legends, 方便 AI 知道你的详细结果涉及哪些知识点, 更好的规划前言框架。FIGURE 1 IFNA is a broad marker of STING pathway activation in NSCLC. (A) An illustration of the various factors essential for STING pathway activation. (B) Basal STING expression in 11 NSCLC cellines detected using western blot. Vinculin was used as a loading control. (C) RNA sequencing data from six NSCLC cell lines demonstrating the expression levels of the 50 genes with the average highest log2 fold change, six hours after transfection with HT-DNA

Data presented as heatmap with Z-scores. (D) A heatmap of IFNL1, IFNL2, and IFNB1 gene expression six hours post stimulation with HT-DNA

measured by RT-qPCR, in eight NSCLC cell lines. (E, F) Evaluation of STING pathway activation in four NSCLC cellines and the epithelial control celline, Nuli-1, following transfection with HT-DNA

or control for six hours. The expression of total STING, phospho(Ser366)STING, total TBK1, phospho (Ser172)TBK1, total IRF3, and phospho(Ser366)IRF3 were evaluated by western blot using vinculin as loading control. (G-1) NSCLC cellines and Nuli-1 were stimulated with HT-DNA

formulated with lipofectamine 2000. Supernatants were harvested after 20 hours and analyzed for IFN and IFN expression by ELISA. In (G, H) data is shown as mean standard deviation of three independent experiments conducted in triplicates. In (I) mean of one experiment performed in duplicates is shown with standard deviation indicated.

FIGURE 2 IFN and IFN share similar cell type dependent kinetic patterns. NSCLC cell lines and Nuli-1 cells were treated with HT-DNA

formulated with lipofectamine 2000, for the indicated number of hours (hrs), where after gene (A-E) and protein (F-J) expression levels of IFNB1, IFNL1, and IFNL2 were determined. Data is shown as mean

standard deviation of duplicates from one experiment.

FIGURE 3 Doxorubicin and cisplatin induce IFNL1 and IFNB1 transcription. (A) An illustration of chemotherapy-induced STING pathway activation. (B-E) Gene expression of IFNL1 and IFNB1 was determined in NSCLC cell lines after a single-value treatment with doxorubicin or cisplatin for 48 hours. Data is shown as mean standard deviation of triplicates. Two-way ANOVA analysis using Sidak's multiple comparisons test is performed. Once again H596 stood out being the only cell line demonstrating significant upregulation of only IFNB1 and not IFNL1 expression (Figure 3C).

FIGURE 4 Doxorubicin and cisplatin induce IFNL1 and IFNB1 transcription in a STING-dependent manner. (A) Western blot of STING expression measured in wild type (WT) or STING knockout (KO) of our IFNCLC cell lines. Vinculin is used as loading control. (B-I) Mock

and STING KO NSCLC cell lines were stimulated with doxorubicin (B-E) or cisplatin (B-F) for 48 hours. Cells were then harvested and gene expression of IFNL1 and IFNB1 determined by RT-qPCR. Data is shown as mean standard deviation of triplicates. Two-way ANOVA analysis using Snak's multiple comparisons test is performed.

FIGURE 5 IFNL1 expression is cell type dependent and is differentiated in tumor cells. (A) Gene expression as normalization transcripts per million (nTMP) of IFNL1, IL10RB, IFNAR1, and IFNAR2 expression in different human tissues generated from the Human Protein Atlas (proteinatlas.org) (59). (B) IFNL1 gene expression as nTMP in different tissues generated from the Human Protein Atlas (proteinatlas.org) (65). (C) The expression of IFNL1 in healthy and malignant epithelial cells obtained from 316 patients (healthy controls or lung cancer patients) using single cell RNA sequencing (42). The difference in expression between the groups was evaluated by a Wilcoxon Rank Sum test.

. (D) The basal gene expression of IFNL1 in four NSCLC cell lines and Nuli-1 cells measured by RT-qPCR. Values are shown as mean standard deviation from 7-5 replicates from one or two experiments. (E) Two NSCLC cell lines were treated for 15 min. with either IFN, (100ng/mL), IFN, (100ng/mL) or left untreated prior to cell lysis. The protein expression of STAT1 and pSTAT1 was detected by western blot. Vinculin was used as loading control.

FIGURE 6 Transcriptional upregulation of IFNL1 increases IFN, sensitivity and leads to decreased viability. (A) Illustration of CRISPR activation (CRISPRa) of IFN1. Created using Biorender.com. (B) The H1650 cell line was subjected to CRISPRa of IFN1 and gene expression evaluated at day 1-4 after treatment. The IFN1R1 expression was normalized to H1650 mock control cells in which transcriptional activation was not performed. The data is shown as mean standard deviation of two replicates from a single experiment. (C) The H1650 cell line was subjected to CRISPRa of IFN1R1 (IFN1R1). At the indicated days, cells were treated with IFN1 or left untreated for

. prior to harvest. The level of STAT1 and pSTAT1 expression was determined by western blot. H1650 mock control cells in which transcriptional activation was not performed were included for each time point. Vinculin was used as a loading control. (D-G) Viability measurement of H1650 (D, E) and H358 (F, G) cell lines. One day after CRISPRa of IFN1R1 and seeding, cells were treated with the indicated concentrations of IFN1 or IFN1 for 48 hours. Viability was compared to an untreated control for both IFN1R1+ and mock cells. Data is shown as mean standard deviation (shaded area) of the average of four replicates from two-three experiments in (D-E, G) In (F) mean standard deviation of four replicates are shown. Two-way ANOVA analysis using Sidak's multiple comparisons test is performed comparing mock and IFN1R1+ in samples treated with of IFN1 or IFN1.

</Figure legend>

<Aims and Scope of target journal>

{{目标期刊的发文范围介绍，方便 AI 根据受众规划侧重点:

Frontiers in Immunology is a multidisciplinary journal that publishes research across basic, translational, and clinical immunology. The journal showcases advances and novel approaches to diagnosing and treating immune disorders in both animal and cellular models,

as well as in humans.

Frontiers in Immunology welcomes contributions that investigate the basic mechanisms of immune system development and function, with a particular emphasis on describing the clinical and immunological phenotype of human immune disorders and defining their molecular basis.}}

</Aims and Scope of target journal>

output format

<example>

1 cGAS-STING 通路在抗肿瘤免疫中的核心地位（200 字）

文献检索关键词：cGAS-STING pathway, tumor immunity, DNA damage response, innate immunity cancer;

内容：阐述 cGAS-STING 作为胞质 DNA 感知通路的基本机制，其在肿瘤免疫监视中的关键作用。

2 I 型与 III 型干扰素的差异化功能（250 字）

文献检索关键词：type I interferon, type III interferon, IFN-lambda, epithelial immunity, antiviral response;

内容：对比 I 型和 III 型 IFN 的表达模式、受体分布和生物学功能，强调 III 型 IFN 的组织特异性。

.....

Research Gap 与研究目标（150 字）

明确 gap：尽管 cGAS-STING 通路广受关注，但其激活后优先诱导 III 型而非 I 型 IFN 的现象未被充分认识；IFNLR1 在肿瘤细胞中的表达调控及其作为免疫逃逸机制的作用尚不清楚。

引出本研究的核心问题和创新点。

</example>

Requirements

1 Introduction 部分总计 800 字，列出每一部分字数规划。

2 列出每个主题的文献综述检索关键词；

3 整个框架的暗线是要突出我的研究的重要性；

4 research gap 要明确具体。

请一步一步思考。

说明：

1 模型：Claude opus 4.1 或 Gemini 2.5 pro；

2 标黄部分换成你自己的内容即可。

2 文献综述调研提示词

请按照如下框架，帮我进行系统、深入、批判性的学术论文文献综述。该文献综述将用于指导我撰写我论文的 Introduction 部分，投稿到 [Frontiers in Immunology](#)。

内容要求

- 优先引用近五年发表在 top 学术期刊的文献。
- 按照我给出的框架逐段展开，不要偏离。
- 字数目标：2000 字左右。

写作风格

- 深入、客观、正式的学术风格

<framework>

上一步生成的前言框架。字数删去，保留 research gap，研究目标删去。例如：

从 DNA 损伤响应到免疫激活的递进式框架

1 DNA 损伤与肿瘤免疫的交互作用

文献检索关键词：DNA damage response, tumor immunity, chemotherapy-induced immunogenicity, damage-associated molecular patterns

内容：阐述肿瘤治疗中 DNA 损伤如何转化为免疫激活信号，化疗药物诱导的免疫原性细胞死亡机制

2 cGAS-STING 通路：连接 DNA 损伤与干扰素响应的桥梁

文献检索关键词：cGAS-STING pathway, cytosolic DNA sensing, interferon response, tumor microenvironment

内容：详述 cGAS-STING 识别胞质 DNA、激活下游信号的分子机制，在肿瘤免疫监视中的核心作用

3 III 型干扰素的独特生物学特性

文献检索关键词：type III interferon, IFN-lambda, epithelial immunity, tissue-specific immunity, IFNLR1 expression

内容：强调 IFN λ 的上皮细胞特异性表达和受体分布的局限性，与 I 型 IFN 的功能差异

4 肿瘤细胞的干扰素逃逸机制

文献检索关键词：tumor immune evasion, interferon resistance, receptor downregulation, NSCLC immunosuppression

内容：综述肿瘤细胞通过调控干扰素受体表达逃避免疫监视的策略

Research Gap

明确 gap：虽然 I 型 IFN 在抗肿瘤免疫中备受关注，但 STING 激活后优先诱导 III 型 IFN 的现象及其生物学意义尚不清楚；IFNLR1 下调作为肿瘤免疫逃逸新机制未被充分认识

</framework>

说明：

1 这一步用各大模型的 deep research 完成，Claude，Gemini，OpenAI 都可以。可以同时使用，综合成一份更完整的综述。

2 字数部分，可以是你的前言字数的 2-3 倍，以便综述更全面。

3 标黄部分换成你自己的内容。

3 前言全文撰写提示词

Role

I am an expert in Immunology with a profound academic background in Immunology and I'm familiar with the professional concepts and terminology in this field. I also have extensive experience in writing academic papers, particularly for the 'Frontiers in Immunology' journal.

Attention

Create a strong Introduction for my article which will be submitted to the journal 'Frontiers in Immunology', emphasizing the significance and contributions of the research.

Goals

Compose an Introduction that meets the quality standards of 'Frontiers in Immunology' based on the designated title and framework of the Introduction section.

Background

<Title>

{{IFNλ1 is a STING-dependent mediator of DNA damage and induces immune activation in lung cancer}}

</Title>

<Abstract>

{{Abstract Introduction: The importance of the cGAS-STING pathway and type I interferon (IFN) in anti-tumor immunity has been widely studied. However, there is limited knowledge about the role of type III IFNs in cancer settings. Type III IFNs, comprising IFNλ1-4, are opposite to type I IFN only expressed by a few cell types, including epithelial cells, and the receptor subunit IFNLR1, is equally only expressed on limited types of cells. Methods: Gene and protein expression of the cGAS-STING signaling pathway was characterized in a series of non-small cell lung cancer (NSCLC) cell lines. Herring-testis DNA stimulation and chemotherapy drugs (doxorubicin and cisplatin) were used to activate the cGAS-STING pathway, and the level of activation was determined by measuring changes in the transcriptomic profile as well as type I and III IFNs by ELISA. Re-expression of IFNLR1 on cancer cell lines was achieved using CRISPR activation (CRISPRa) followed by evaluating chemotherapy-induced apoptosis using flow cytometry assays. Results: STING was not broadly expressed across the NSCLC cell lines. Those cancer cell lines expressing all relevant factors supporting the cGAS-STING pathway secreted IFNλ following STING activation whereas only few of them expressed IFNβ. Treatment with chemotherapy drugs likewise preferentially induced IFNλ, which was abrogated in CRISPR-Cas9 STING knock-out cells. Expression of IFNLR1 was found downregulated in the cancer cell lines compared to the benign epithelial cell line Nuli-1. Rescuing IFNLR1 expression by CRISPRa in multiple cancer cell lines sensitization them to IFNλ-stimulation and resulted in significant reduction in cell viability. Conclusion: Downregulation of IFNLR1 can be an immune evasion mechanism developed by cancer cells to avoid responding to endogenous type III IFNs. Thus, rescuing

IFNLR1 expression in NSCLC in conjunction to chemotherapy may potentially be harnessed to elevate the anti-tumoral responses.}}

</Abstract>

<framework>

1 DNA 损伤与 cGAS-STING 通路在肿瘤免疫中的核心作用 (约 150 字)

2 I 型与 III 型干扰素的差异性表达及功能特征 (约 180 字)

3 III 型干扰素在肿瘤微环境中的独特作用 (约 150 字)

4 IFNLR1 在肿瘤中的表达调控与免疫逃逸机制 (约 200 字)

5 Research Gap 与本研究目标 (约 120 字)

</framework>

Skills

Establishing Context and Rationale:

- Start with a broad overview of the central topic or issue to provide background context.
- Progressively narrow down the focus to highlight limitations in current knowledge and gaps in the literature.
- Conclude this section by clearly stating the specific research problem or question the current study aims to address.
- Emphasize the real-world significance, relevance, timeliness, and importance of the research throughout.

Structure and Logical Flow:

- Organize content to start broad, providing a wide lens on the research area, then funnel down to the specifics of the current study.
- State the research objectives, knowledge gaps, hypotheses, and questions early in the Introduction.
- Ensure each paragraph builds upon the previous one in a logical progression, establishing connections between ideas.

Style, Tone and Wording:

Maintain a formal yet accessible scholarly tone suited to the target journal and audience.

Use clear, simple, concise sentences favoring active voice. Avoid complex jargon.

Vary sentence structure for enhanced readability. Use transitions to link ideas.

Define specialized terms on first use. Summarize relevant prior research briefly.

Support key points with evidence from the literature. Demonstrate command of the research area.

Engagement and Transitions:

Engage readers from the start by providing detailed background on the research problem.

Use transitions and topic sentences to guide the reader through the logical flow.

Vary sentence length and structure to aid readability and engagement.

Precision and Clarity:

Remain sharply focused on the research problem and aims. Avoid extraneous details.
Explicitly state the research gaps, questions, and objectives early on.
Strive for precision, clarity, and concision in articulating the research goals and rationale.

End Strongly:

Conclude by underscoring the significance, relevance, and expected contributions of the study.

End with a clear statement of research aims and importance to orient the reader.

Example

<Title>

{{SLAMF7 (CD319) enhances cytotoxic T-cell differentiation and sensitizes CD8(+) T cells to immune checkpoint blockade}}

</Title>

<Introduction>

{{Restoring effective T-cell responses is central to tumor immunosurveillance and cancer control. Tumors employ immune escape mechanisms - including secretion of TGF- β and expression of PD-L1 - to inhibit CD8(+) T-cell effector function (1). Immune checkpoint blockade (ICB) with antibodies targeting CTLA-4 or PD-1/PD-L1 can reverse this suppression and reinvigorate anti-tumor immunity (2). ICB has become a key component of treatment across multiple cancer types. In head and neck squamous cell carcinoma (HNSCC), PD-1 inhibitors such as Pembrolizumab and Nivolumab are approved as first-line therapies for recurrent or metastatic disease and combinational therapy is arising (3–5). However, therapeutic efficacy of ICB in boosting tumor rejection is limited by immune-related adverse events (e.g., colitis, hepatitis, pneumonitis, endocrinopathies) and low rates of durable responses (6–8). Accordingly, there is a pressing need to develop next-generation immunotherapies that extend clinical benefit and limit treatment-associated toxicity.

Analysis of downstream targets of the inhibitory molecule CTLA-4 on CD8(+) T cells revealed potential targets to improve or provide an alternative to established immune checkpoint therapies (9). One of the potential targets is the self-ligating receptor SLAMF7 (CD319), a member of the Signaling Lymphocyte Activation Molecule (SLAM) family expressed on hematopoietic cells (10–15). Through its cytoplasmic immunoreceptor tyrosine-based switch motifs (ITSMs), it has been proposed to recruit several Src homology-2 (SH-2) domain-containing adapter proteins such as EAT-2, SHP-1 and SHP-2, SHIP1, Csk, Fyn or PLC- γ (14–16). On NK cells, SLAMF7 is known to promote activation and degranulation, but also mediate inhibitory signals in the absence of the adaptor EAT-2 (13, 14, 16). Due to its high expression on myeloma cells monoclonal antibodies have been developed to target SLAMF7 as a tumor antigen, leading to NK-cell mediated antibody-dependent cellular cytotoxicity (ADCC) (17–19). Further it has been shown that SLAMF7 promotes phagocytosis of cancer cells by macrophages through interaction with Mac-1 (20).

In T cells, SLAMF7 is predominantly expressed on cytotoxic CD8(+) T cells and on a subset of cytolytic CD4(+) T cells (21, 22). Its engagement has been shown to restore effector function in dysfunctional CD8(+) T cells in systemic lupus erythematosus, and to enhance cytotoxicity in tumor-specific CD4(+) T cells upon agonistic stimulation (23, 24). However, in cancer-

related contexts, SLAMF7 expression on CD8(+) T cells is also associated with T cell exhaustion and a suppressive phenotype (25, 26). In a murine model, SLAMF7 engagement during CD8(+) T-cell priming was shown to integrate environmental cues into acquisition of cytotoxic effector functions (15). These findings underline the ambivalent role of SLAMF7 in T-cell biology, ranging from immune activation to dysfunction, depending on the cellular and environmental context.

In this study, we aimed to resolve this ambiguity by identifying upstream signals that regulate SLAMF7 expression in human CD8(+) T cells and by assessing whether SLAMF7 co-stimulation enhances or impairs CD8(+) T-cell responses. Our findings provide insights into the contextual role of SLAMF7 in CD8(+) T cells and explore its potential as a target to modulate T-cell function for cancer immunotherapy.}}

</Introduction>

Constraints

- Adhere strictly to the provided framework.
- Limit your writing strictly to the Introduction section only.
- In-text citations in the literature review materials (e.g., "(Zhang, 2025)" or numbered style "[1,2]") must be preserved exactly as they appear.
- The content in the ## Example section is provided solely to illustrate the structural and stylistic conventions of an academic Introduction. Do not confuse or conflate the example's case content with the scientific focus described in my ## Background section.

Workflows:

- 1 Carefully review the attached literature review materials. Based on my research topic, goals, and findings, determine which parts of the review to retain, revise, or omit — and clearly justify all additions or deletions.
- 2 Plan the paragraph structure and approximate word count for each paragraph in the Introduction, emphasizing the significance of this study.
- 3 Draft a complete 800-word first version of the Introduction.
- 4 Study the stylistic features of the ## Example section and apply the academic writing techniques listed in the ## Skills section. Use these to revise the initial draft into a more formal, neutral, and academically polished version. Ensure varied sentence lengths, smooth logical flow, and coherent transitions. Any newly added factual claims that require citations should be clearly marked with "(citation needed)" as a reminder for later completion.

Please think step by step.

-
- 1 模型：Claude opus 4.1;
 - 2 标黄部分换成自己的内容，只替换{{ }}里面的内容;
 - 3 ## Example 部分给一篇你想要模仿的例文，让 AI 更好的学习文风。
 - 4 文献综述部分单独粘贴。